

Pulling the app from github

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder
$ git clone https://github.com/dockerexamples/example-voting-app.git
Cloning into 'example-voting-app'...
remote: Enumerating objects: 1179, done.
remote: Total 1179 (delta 0), reused 0 (delta 0), pack-reused 1179 (from 1)
Receiving objects: 100% (1179/1179), 1.21 MiB | 1.64 MiB/s, done.
Resolving deltas: 100% (448/448), done.

Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder
```

Deploying locally

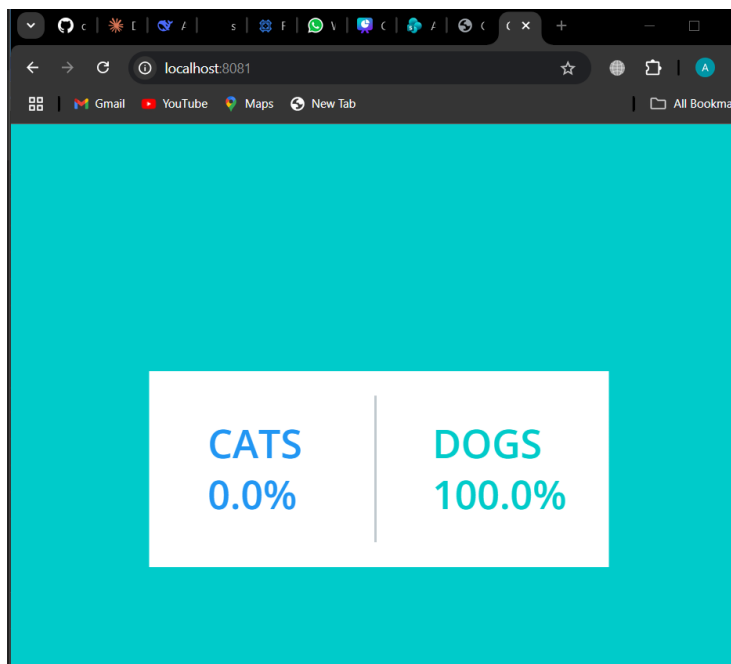
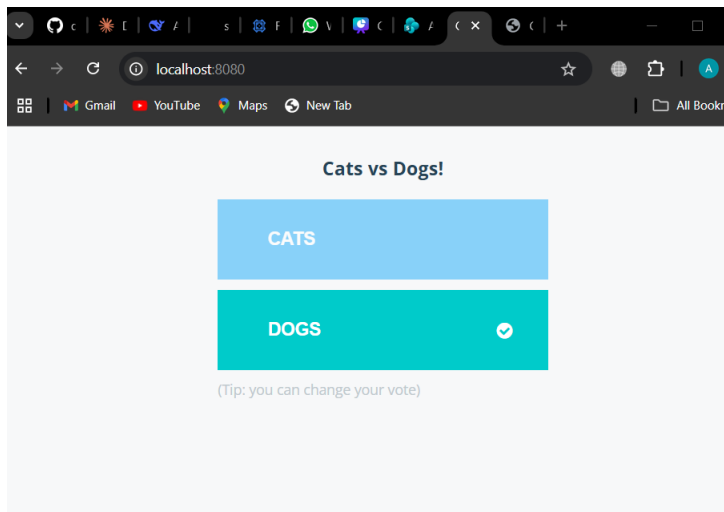
```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ docker-compose up -d
[+] Running 20/20
  ✓ redis Pulled                                45.5s
  ✓ cffa735aaa17 Pull complete                  1.4s
  ✓ 2d35ebdb57d9 Pull complete                  7.8s
  ✓ 362eff98b603 Pull complete                 39.5s
  ✓ 793133e9ba6b Pull complete                 39.8s
  ✓ c84a6e003d26 Pull complete                  1.6s
  ✓ 4f4fb700ef54 Pull complete                  0.1s
  ✓ 6a229eaa0f8a Pull complete                  8.3s
  ✓ db Pulled                                  71.4s
  ✓ 25fbe13bb576 Pull complete                  1.7s
  ✓ 141979064465 Pull complete                  1.7s
  ✓ 9fa3a17171bc Pull complete                 65.3s
  ✓ 86a96880bc80 Pull complete                  1.1s
  ✓ 1074353eec0d Pull complete                 17.5s
  ✓ 1de01a2b564f Pull complete                  2.7s
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ docker-compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
example-voting-app-db-1	postgres:15-alpine	"docker-entrypoint.s..."	db	16 minutes ago	Up 16 minutes (health)
example-voting-app-redis-1	redis:alpine	"docker-entrypoint.s..."	redis	16 minutes ago	Up 16 minutes (health)
example-voting-app-result-1	example-voting-app-result	"nodemon --inspect=0..."	result	16 minutes ago	Up 16 minutes
example-voting-app-vote-1	example-voting-app-vote	"python app.py"	vote	16 minutes ago	Up 16 minutes (health)
example-voting-app-worker-1	example-voting-app-worker	"dotnet Worker.dll"	worker	16 minutes ago	Up 16 minutes

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
```



Working.. Stopping local services

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ docker-compose down
[+] Running 7/7
  ✓ Container example-voting-app-vote-1      Removed      1.9s
  ✓ Container example-voting-app-worker-1    Removed      0.8s
  ✓ Container example-voting-app-result-1    Removed      2.3s
  ✓ Container example-voting-app-redis-1     Removed      0.8s
  ✓ Container example-voting-app-db-1        Removed      0.7s
  ✓ Network example-voting-app_front-tier    Removed      0.8s
  ✓ Network example-voting-app_back-tier     Removed      1.6s
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
```

Building and pushing images to Docker Hub:

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ docker build -t arsal7477/vote-app ./vote
[+] Building 5.2s (12/12) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 1.12kB                             0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 3.6s
=> [auth] library/python:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [base 1/5] FROM docker.io/library/python:3.11-slim@sha256:158caf0e080e2cd74ef2879ed3c4e697792ee65251c8208b7afb56683c32e 0.1s
=> => resolve docker.io/library/python:3.11-slim@sha256:158caf0e080e2cd74ef2879ed3c4e697792ee65251c8208b7afb56683c32e 0.1s
=> [internal] load build context                                  0.1s
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ docker build -t arsal7477/result-app ./result
[+] Building 3.5s (13/13) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 550B                                 0.0s
=> [internal] load metadata for docker.io/library/node:18-slim   2.6s
=> [auth] library/node:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 55B                                       0.0s
=> [1/7] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651838830e 0.1s
=> => resolve docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651838830e 0.1s
=> [internal] load build context                                  0.0s
=> => transferring context: 593B                                       0.0s
=> CACHED [2/7] RUN apt-get update && apt-get install -y --no-install-recommends curl tini && rm -rf /var/lib/apt/ 0.0s
=> CACHED [3/7] WORKDIR /usr/local/app                           0.0s
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ docker build -t arsal7477/worker-app ./worker
[+] Building 2.9s (16/16) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 1.13kB                             0.1s
=> [internal] load metadata for mcr.microsoft.com/dotnet/sdk:7.0 2.0s
=> [internal] load metadata for mcr.microsoft.com/dotnet/runtime:7.0 2.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [stage-1 1/3] FROM mcr.microsoft.com/dotnet/runtime:7.0@sha256:8cd26108e7a329458e5b84342816ae39b2b38bd041cfdc0b79b8cf1c 0.2s
=> => resolve mcr.microsoft.com/dotnet/runtime:7.0@sha256:8cd26108e7a329458e5b84342816ae39b2b38bd041cfdc0b79b8cf1c0e0d376f 0.2s
=> [build 1/7] FROM mcr.microsoft.com/dotnet/sdk:7.0@sha256:d32bd65cf5843f413e81f5d917057c82da99737cb1637e905a1a4bc2e7ec6c 0.2s
=> => resolve mcr.microsoft.com/dotnet/sdk:7.0@sha256:d32bd65cf5843f413e81f5d917057c82da99737cb1637e905a1a4bc2e7ec6c8d 0.2s
=> [internal] load build context                                  0.1s
=> => transferring context: 95B                                       0.0s
=> CACHED [stage-1 2/3] WORKDIR /app                             0.0s
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ docker push arsal7477/vote-app
Using default tag: latest
The push refers to repository [docker.io/arsal7477/vote-app]
afd237575c62: Pushed
bb40365316ee: Pushed
1733a4cd5954: Pushed
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ docker push arsal7477/result-app
Using default tag: latest
The push refers to repository [docker.io/arsal7477/result-app]
f45e11c7e3d1: Pushed
2b38783cb300: Pushed
8c994cf49dd1: Pushed
b1831021e35a: Pushed
61320b01ae5e: Pushed
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ docker push arsal7477/worker-app
Using default tag: latest
The push refers to repository [docker.io/arsal7477/worker-app]
38d12d6ca965: Pushed
82bb7a80de57: Pushed
728328ac3bde: Pushed
```

STEP 2: AZURE RESOURCE GROUP CREATION

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ az login
Select the account you want to log in with. For more information on login with Azure CLI, see https://go.microsoft.com/fwlink/?lin
kid=2271136

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] *   Azure for Students   7
The default is marked with an *: the default tenant is 'Higher Education Commission' and subscription is 'Azure for Students' (706
```

Create resource group

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project
/New folder/example-voting-app (main)
$ az group create --name voting-app-rg --location eastus
{
  "id": "/",
  "locatio
  "managed
  "name":
  "properl
  "provi
  },
  "tags":
  "type":
}
```

Verification

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Resource Manager

Resource Manager | Resource groups

How to manage changes with deployment tools? Summarize my costs by service Export resource groups using Bicep or Terraform

Search

+ Create Manage view Refresh Export to CSV Open query Assign tags Add to service group

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

You are viewing a new version of Browse experience. Click here to access the old experience.

Filter for any field... Subscription equals all Location equals all Add filter

☐	Name ↑	Subscription	Location
☐	voting-app-rg	Azure for Stude...	East US

STEP 4: CREATE AKS CLUSTER

Home > Kubernetes center (preview) | Clusters >

Create Kubernetes cluster ...

Recommend VM sizes for Kubernetes Service

Recommend VM sizes for Kubernetes Service for my AI workloads

BasicsNode poolsNetworkingIntegrationsMonitoringSecurityAdvancedTagsReview + create

[View automation template](#)**Basics**

Subscription	Azure subscription 1
Resource group	voting-app-rg
Region	Southeast Asia
Kubernetes cluster name	voting-aks-cluster
Kubernetes version	1.33.5
Automatic upgrade	patch
Automatic upgrade scheduler	Every week on Sunday (recommended)
Node security channel type	NodeImage
Security channel scheduler	Every week on Sunday (recommended)

Node pools

Node pools	1
Enable virtual nodes	Disabled
Node Auto-provisioning	Disabled

Previous

Next

Create

Home >

microsoft.aks-1766164701955 | Overview

...

Deployment

Search

×

◀

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name: microsoft.aks-1766164701955

Subscription: Azure subscription 1

Resource group: voting-app-rg

Start time: 12/19/2025, 10:19:24 PM

Correlation ID: 005a29bc-cba1-4add-8739-3e8e489272f3

Deployment details

Resource	Type	Status	Operation details
No results.			

Give feedback

[Tell us about your experience with deployment](#)

Home > **microsoft.aks-1766164701955 | Overview** ✕

Deployment

» Delete Cancel Redeploy Download Refresh

✓ **Your deployment is complete**

Deployment name: microsoft... Start time: 12/19/2025, 10:19:24 PM
 Subscription: [Azure subscript...](#) Correlation ID: 005a29bc-cba1-4add-87
 Resource group: [voting-app-...](#)

Deployment details

Resource	Type	Status
----------	------	--------

Cost Management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)

portal.azure.com/?l=en-us#@arsalanswat7@gmail531.onmicrosoft.com/resource/subscriptions/bcb59c26-1db3-414f-8d58-d02d30c5de6e/resourceGroups/voting-app-r...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > **voting-aks-cluster** ✕

Kubernetes service

Search

Overview

Activity log

Access control (IAM)

Tags

Monitor

Diagnose and solve problems

Microsoft Defender for Cloud

Cost analysis

Resource visualizer

Kubernetes resources

Settings

Monitoring

Automation

Help

Create Connect Start Stop Delete Refresh Open in mobile Give feedback

Essentials

Resource group	: voting-app-rg	Kubernetes version	: 1.33.5
Power state	: Running	API server address	: voting-aks-cluster-dns-giudbllq.hcp.southeastasia.azmk8s.io
Cluster operation status	: Succeeded	SKU	: Base
Subscription	: Azure subscription.1	Pricing tier	: Free
Location	: Southeast Asia	Network configuration	: Azure CNI Overlay
Subscription ID	: bcb59c26-1db3-414f-8d58-d02d30c5de6e	Node pools	: 1 node pool
Fleet Manager	: Click here to assign	Container registries	: Attach a registry
		Created time	: December 19, 2025 at 10:19 PM

Tags (edit) : [Add tags](#)

Get started **Properties** Monitoring Recommendations

Kubernetes services

Encryption type	Encryption at-rest with a platform-managed key
Virtual node pools	Not enabled

Node pools

Node pools	1 node pool
Kubernetes versions	1.33.5
Node sizes	

Networking

API server address	voting-aks-cluster-dns-giudbllq.hcp.southeastasia.azmk8s.io
Network configuration	Azure CNI Overlay
Pod CIDR	10.244.0.0/16
Service CIDR	10.0.0.0/16
DNS service IP	10.0.0.10
Cilium dataplane	Not enabled
Network policy engine	None

Add or remove favorites by pressing Ctrl+Shift+F

Node created

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
aks-testnode-30486093-vms1         Ready    <none>    5m9s   v1.33.5
```

Deploy the Voting Application

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and D
distributed Computing/project/New folder/example-voting-app (main)
$ kubectl apply -f k8s-specifications/
E1219 23:45:39.601789 17912 request.go:1332] Unexpected error when reading response body: context de
dline exceeded (Client.Timeout or context cancellation while reading body)
deployment.apps/db created
service/db created
deployment.apps/redis created
service/redis created
deployment.apps/result created
service/result created
deployment.apps/vote created
service/vote created
deployment.apps/worker created
```

Workloads before deploying the application

Home > voting-aks-cluster

voting-aks-cluster | Workloads ☆

Kubernetes service

» + Create ▾ ↗ Scale 🗑 Delete ↻ Refresh 🏷 Show labels 🗨 Give feedback

Deployments Pods Replica sets Stateful sets Daemon sets Jobs Cron jobs

Filter by namespace
All namespaces ▾ Deployment name : All ▾ Add label filter

☐	Name	Namespace	Ready	Age ↓	CPU	Me
☐	azure-wi-webhook-con	kube-system	✔ 2/2	1 hour	🔥 Enable metrics	
☐	eraser-controller-man	kube-system	✔ 1/1	1 hour	▬ %	▬ %
☐	coredns	kube-system	✔ 2/2	1 hour	▬ %	▬ %
☐	coredns-autoscaler	kube-system	✔ 1/1	1 hour	▬ %	▬ %
☐	konnectivity-agent	kube-system	✔ 2/2	1 hour	▬ %	▬ %
☐	konnectivity-agent-aut	kube-system	✔ 1/1	1 hour	▬ %	▬ %
☐	metrics-server	kube-system	✔ 2/2	1 hour	▬ %	▬ %

Workloads after deploying the application

Microsoft Azure Upgrade 🔍 Search resources, services, and docs (G+7) Copilot

Home > voting-aks-cluster

voting-aks-cluster | Workloads ☆

Kubernetes service

🔍 Search « + Create ▾ ↗ Scale 🗑 Delete ↻ Refresh 🏷 Show labels 🗨 Give feedback

Overview Activity log Access control (IAM) Tags Monitor Diagnose and solve problems Microsoft Defender for Cloud Cost analysis Resource visualizer Kubernetes resources Namespaces Workloads Services and ingress Network policies Storage Configuration Custom resources Events

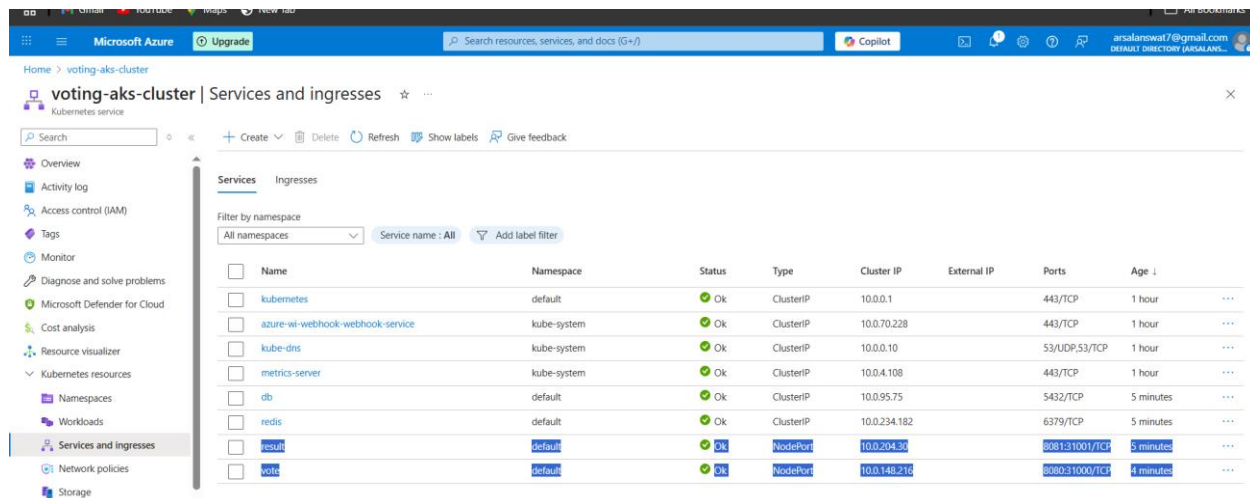
Deployments Pods Replica sets Stateful sets Daemon sets Jobs Cron jobs

Filter by namespace
All namespaces ▾ Deployment name : All ▾ Add label filter

☐	Name	Namespace	Ready	Age ↓	CPU	Memory
☐	azure-wi-webhook-controller-manager	kube-system	✔ 2/2	1 hour	🔥 Enable metrics	
☐	eraser-controller-manager	kube-system	✔ 1/1	1 hour	▬ %	▬ %
☐	coredns	kube-system	✔ 2/2	1 hour	▬ %	▬ %
☐	coredns-autoscaler	kube-system	✔ 1/1	1 hour	▬ %	▬ %
☐	konnectivity-agent	kube-system	✔ 2/2	1 hour	▬ %	▬ %
☐	konnectivity-agent-autoscaler	kube-system	✔ 1/1	1 hour	▬ %	▬ %
☐	metrics-server	kube-system	✔ 2/2	1 hour	▬ %	▬ %
☐	db	default	✔ 1/1	45 seconds	▬ %	▬ %
☐	redis	default	✔ 1/1	43 seconds	▬ %	▬ %
☐	result	default	✔ 1/1	42 seconds	▬ %	▬ %
☐	vote	default	✔ 1/1	41 seconds	▬ %	▬ %
☐	worker	default	✔ 1/1	40 seconds	▬ %	▬ %

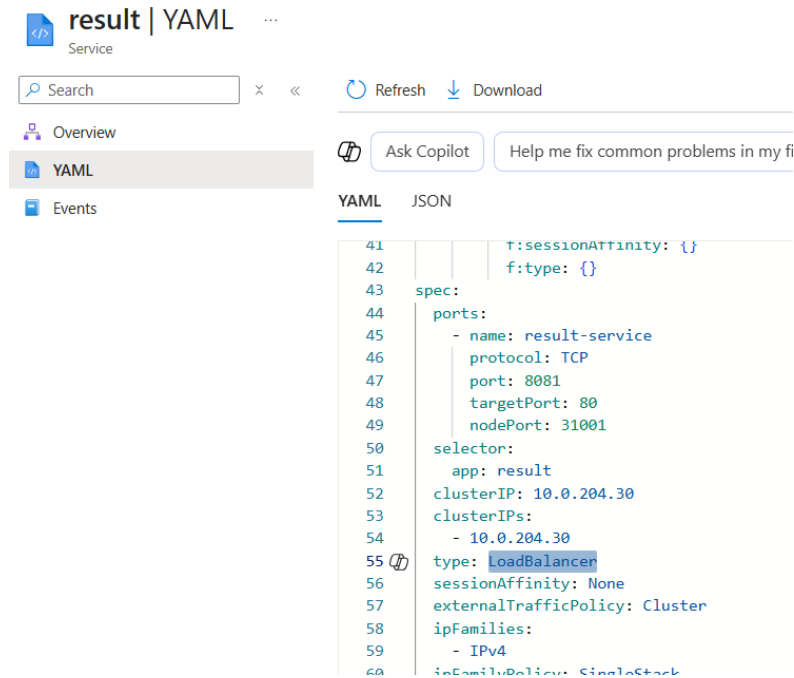
No external IPs means available locally not publicly

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get svc
NAME         TYPE          CLUSTER-IP    EXTERNAL-IP   PORT(S)          AGE
db           ClusterIP     10.0.95.75    <none>        5432/TCP         3m8s
kubernetes   ClusterIP     10.0.0.1      <none>        443/TCP          87m
redis        ClusterIP     10.0.234.182  <none>        6379/TCP         3m6s
result       NodePort      10.0.204.30   <none>        8081:31001/TCP   3m5s
vote         NodePort      10.0.148.216  <none>        8080:31000/TCP   3m4s
```



Name	Namespace	Status	Type	Cluster IP	External IP	Ports	Age
kubernetes	default	Ok	ClusterIP	10.0.0.1		443/TCP	1 hour
azure-wi-webhook-webhook-service	kube-system	Ok	ClusterIP	10.0.70.228		443/TCP	1 hour
kube-dns	kube-system	Ok	ClusterIP	10.0.0.10		53/UDP,53/TCP	1 hour
metrics-server	kube-system	Ok	ClusterIP	10.0.4.108		443/TCP	1 hour
db	default	Ok	ClusterIP	10.0.95.75		5432/TCP	5 minutes
redis	default	Ok	ClusterIP	10.0.234.182		6379/TCP	5 minutes
result	default	Ok	NodePort	10.0.204.30		8081:31001/TCP	5 minutes
vote	default	Ok	NodePort	10.0.148.216		8080:31000/TCP	4 minutes

Changing type of results and votes:



```
41 t:sessionAffinity: {}
42 f:type: {}
43 spec:
44   ports:
45     - name: result-service
46       protocol: TCP
47       port: 8081
48       targetPort: 80
49       nodePort: 31001
50   selector:
51     app: result
52   clusterIP: 10.0.204.30
53   clusterIPs:
54     - 10.0.204.30
55   type: LoadBalancer
56   sessionAffinity: None
57   externalTrafficPolicy: Cluster
58   ipFamilies:
59     - IPv4
60   ipFamilyPolicy: SingleStack
```


Vote

 **vote | YAML** ...

Service

Refresh Download

Overview

YAML

Events

YAML

JSON

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

f:sessionAffinity: {}

f:type: {}

spec:

ports:

- name: vote-service

protocol: TCP

port: 8080

targetPort: 80

nodePort: 31000

selector:

app: vote

clusterIP: 10.0.148.216

clusterIPs:

- 10.0.148.216

type: LoadBalancer

sessionAffinity: None

externalTrafficPolicy: Cluster

Ask Copilot

Help me fix common problems in my file

He

Services

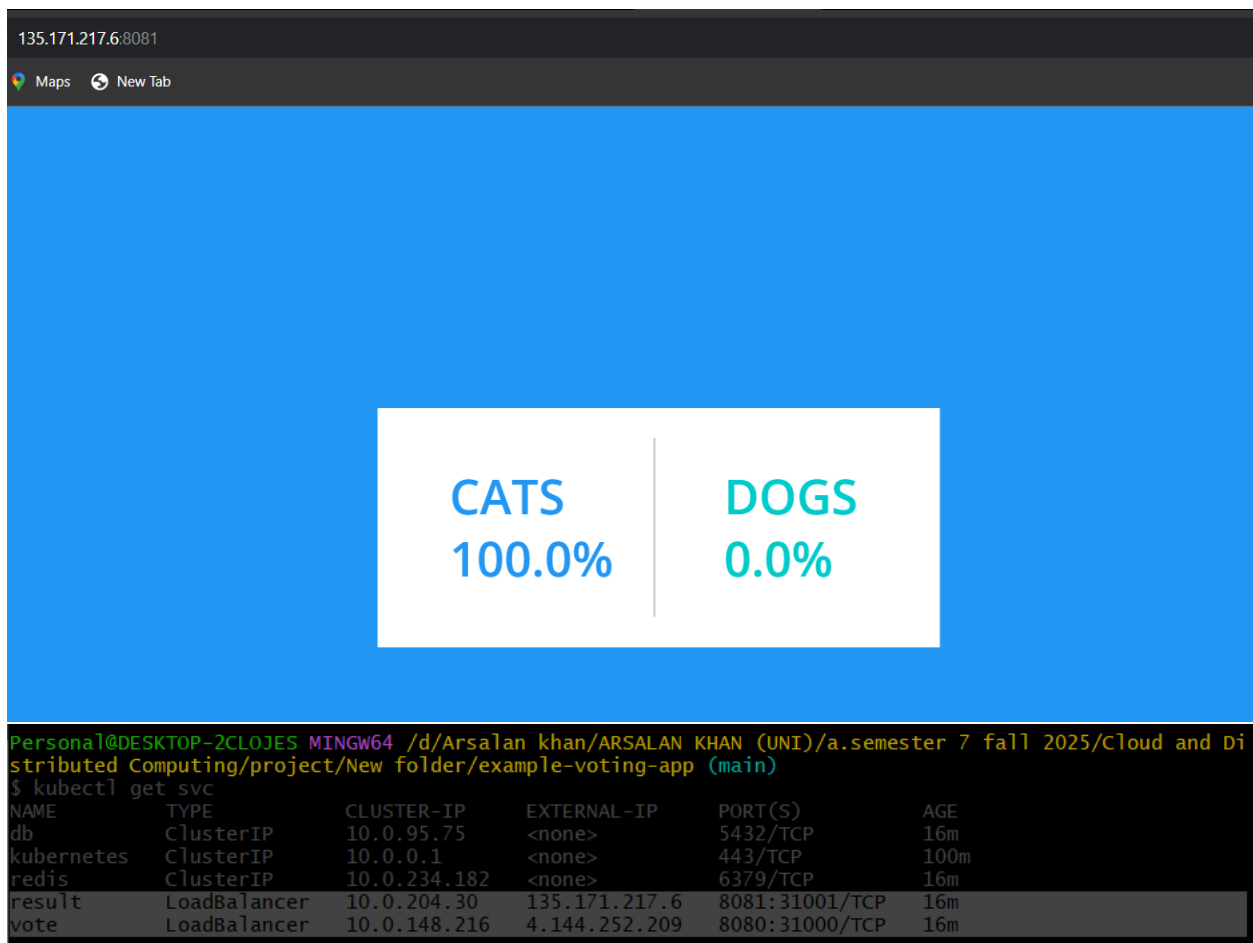
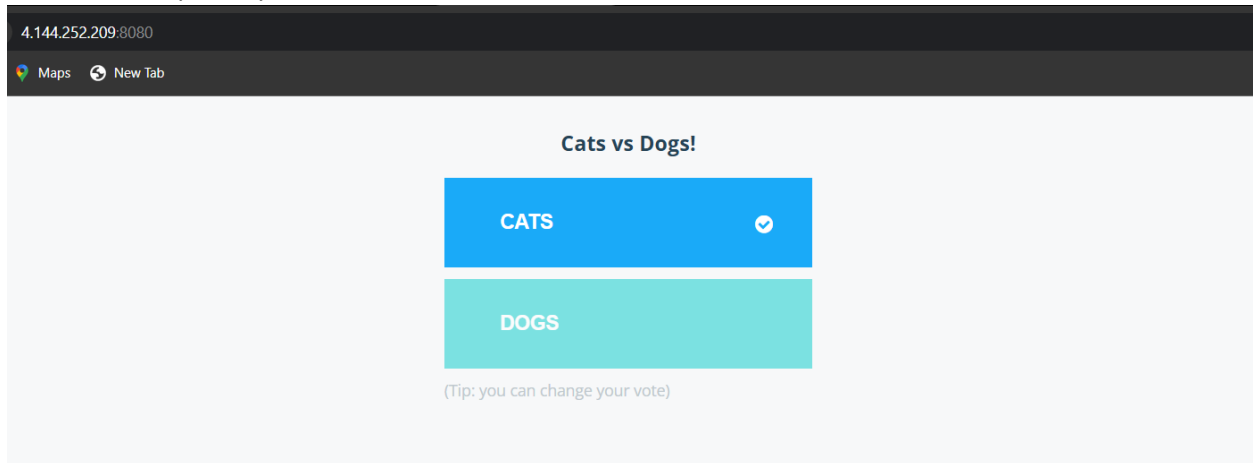
Ingresses

Filter by namespace

All namespaces Service name : All Add label filter

☐	Name	Namespace	Status	Type	Cluster IP	External IP	Ports	Age ↓
☐	kubernetes	default	Ok	ClusterIP	10.0.0.1		443/TCP	2 hours
☐	azure-wi-webhook-webhook-service	kube-system	Ok	ClusterIP	10.0.70.228		443/TCP	2 hours
☐	kube-dns	kube-system	Ok	ClusterIP	10.0.0.10		53/UDP,53/TCP	2 hours
☐	metrics-server	kube-system	Ok	ClusterIP	10.0.4.108		443/TCP	2 hours
☐	db	default	Ok	ClusterIP	10.0.95.75		5432/TCP	14 minutes
☐	redis	default	Ok	ClusterIP	10.0.234.182		6379/TCP	14 minutes
☒	result	default	Ok	LoadBalancer	10.0.204.30	135.171.217.6	8081:31001/TCP	14 minutes
☒	vote	default	Ok	LoadBalancer	10.0.148.216	4.144.252.209	8080:31000/TCP	14 minutes

Now both are publicly available:



```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/
distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get deployments -n default
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
db         1/1     1            1           18m
redis      1/1     1            1           18m
result     1/1     1            1           18m
vote       1/1     1            1           18m
worker     1/1     1            1           18m
```

Implement & Demonstrate Core Requirements

Auto-Scaling (Horizontal Pod Autoscaler - HPA)

downloading the Metrics Server a tool that collects CPU/memory usage from pods, enabling auto-scaling

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Di
distributed Computing/project/New folder/example-voting-app (main)
$ curl -LO https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0 --:--:--  0:00:02 --:--:--    0
  0     0    0     0    0     0      0      0 --:--:--  0:00:03 --:--:--    0
100 4330 100 4330    0     0 676      0  0:00:06  0:00:06 --:--:-- 2360
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Di
distributed Computing/project/New folder/example-voting-app (main)
$ kubectl apply -f components.yaml
E1220 00:17:21.892574 18036 request.go:1332] Unexpected error when reading response body: context
dline exceeded
serviceaccount/metrics-server configured
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader unchanged
clusterrole.rbac.authorization.k8s.io/system:metrics-server configured
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader configured
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator configured
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server configured
service/metrics-server configured
deployment.apps/metrics-server configured
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io configured
```

Autoscaling vote

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Di
distributed Computing/project/New folder/example-voting-app (main)
$ kubectl autoscale deployment vote --cpu-percent=50 --min=2 --max=5
horizontalpodautoscaler.autoscaling/vote autoscaled
```

Result

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Di
distributed Computing/project/New folder/example-voting-app (main)
$ kubectl autoscale deployment result result --cpu-percent=50 --min=2 --max=5
horizontalpodautoscaler.autoscaling/result autoscaled
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Di
distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get hpa
NAME      REFERENCE          TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
result    Deployment/result   cpu: <unknown>/50%  2         5         1          16s
vote      Deployment/vote     cpu: <unknown>/50%  2         5         2          4m30s
```

Load test:

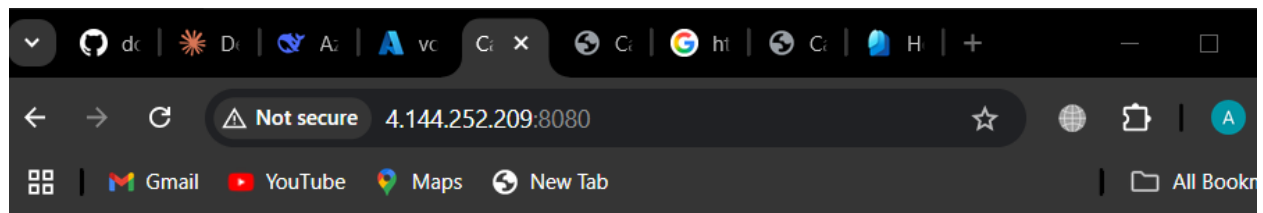
Creating a temporary pod for generating load

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl run load-generator --image=busybox --restart=Never -- /bin/sh -c "while sleep 0.1; do wget -q -O- http://vote; done"
pod/load-generator created
```

Vote service under stress but still working smoothly

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl top pods -n default
NAME                                CPU(cores)   MEMORY(bytes)
db-74574d66dd-hqk6m                 3m           31Mi
redis-6c5fb9c4b7-q7nd1              3m           5Mi
result-5f99548f7c-2qctc             1m           22Mi
result-5f99548f7c-gf55f             1m           22Mi
vote-5d74dcd7c7-49jgh               1m           98Mi
vote-5d74dcd7c7-wzshb               1m           91Mi
worker-6f5f6cdd56-f88ww             13m          49Mi
```

Site was still accessible



Fault tolerance test

Deleting a pod manually

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl delete pod vote-5d74dcd7c7-49jgh -n default
pod "vote-5d74dcd7c7-49jgh" deleted
```

but it self-healed:

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get pods -n default -w
NAME                                READY    STATUS    RESTARTS   AGE
db-74574d66dd-hqk6m                 1/1      Running   0           57m
redis-6c5fb9c4b7-q7nd1              1/1      Running   0           57m
result-5f99548f7c-2qctc             1/1      Running   0           19m
result-5f99548f7c-gf55f             1/1      Running   0           57m
vote-5d74dcd7c7-cxzvf               1/1      Running   0           24s
vote-5d74dcd7c7-wzshb               1/1      Running   0           23m
worker-6f5f6cdd56-f88ww             1/1      Running   0           57m
```

No downtime occurred

5. Load balancing

We have seen the load-balancing above but to explicitly confirm we can see the replicas of vote and result that further confirms the load balancing.

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl top pods -n default
NAME                                CPU(cores)   MEMORY(bytes)
db-74574d66dd-hqk6m                 3m           31Mi
redis-6c5fb9c4b7-q7nd1              3m           5Mi
result-5f99548f7c-2qctc             1m           24Mi
result-5f99548f7c-gf55f             1m           24Mi
vote-5d74dcd7c7-cxzvf               1m           91Mi
vote-5d74dcd7c7-wzshb               1m           95Mi
worker-6f5f6cdd56-f88ww             13m          46Mi

Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get deployments -n default
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
db        1/1     1            1           62m
redis     1/1     1            1           62m
result    2/2     2            2           62m
vote      2/2     2            2           62m
worker    1/1     1            1           62m
```

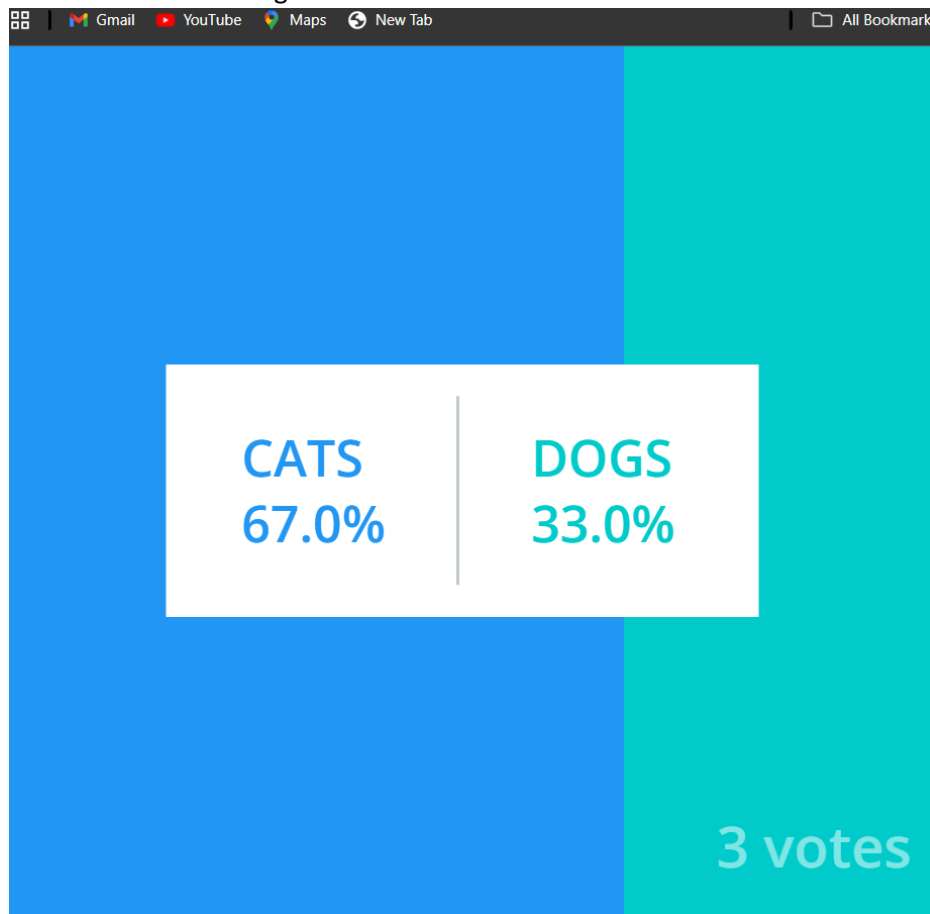
Logs generated for pods during the load testing were almost the same:

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl logs vote-5d74dcd7c7-cxzvf -n default
[2025-12-19 19:42:43 +0000] [1] [INFO] Starting gunicorn 23.0.0
[2025-12-19 19:42:43 +0000] [1] [INFO] Listening at: http://0.0.0.0:80 (1)
[2025-12-19 19:42:43 +0000] [1] [INFO] Using worker: sync
[2025-12-19 19:42:43 +0000] [7] [INFO] Booting worker with pid: 7
[2025-12-19 19:42:43 +0000] [8] [INFO] Booting worker with pid: 8
[2025-12-19 19:42:43 +0000] [9] [INFO] Booting worker with pid: 9
[2025-12-19 19:42:43 +0000] [10] [INFO] Booting worker with pid: 10
```

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl logs vote-5d74dcd7c7-wzshb -n default
[2025-12-19 19:19:42 +0000] [1] [INFO] Starting gunicorn 23.0.0
[2025-12-19 19:19:42 +0000] [1] [INFO] Listening at: http://0.0.0.0:80 (1)
[2025-12-19 19:19:42 +0000] [1] [INFO] Using worker: sync
[2025-12-19 19:19:42 +0000] [8] [INFO] Booting worker with pid: 8
[2025-12-19 19:19:43 +0000] [9] [INFO] Booting worker with pid: 9
[2025-12-19 19:19:43 +0000] [10] [INFO] Booting worker with pid: 10
[2025-12-19 19:19:43 +0000] [11] [INFO] Booting worker with pid: 11
10.224.0.4 - - [19/Dec/2025:19:30:30 +0000] "GET /sitemap.xml HTTP/1.1" 404 207 "-" "Mozilla/5.0 (compatible; CensysInspect/1.1; +https://about.censys.io/)"
```

Data Persistence Test

Casted few vote through different accounts:

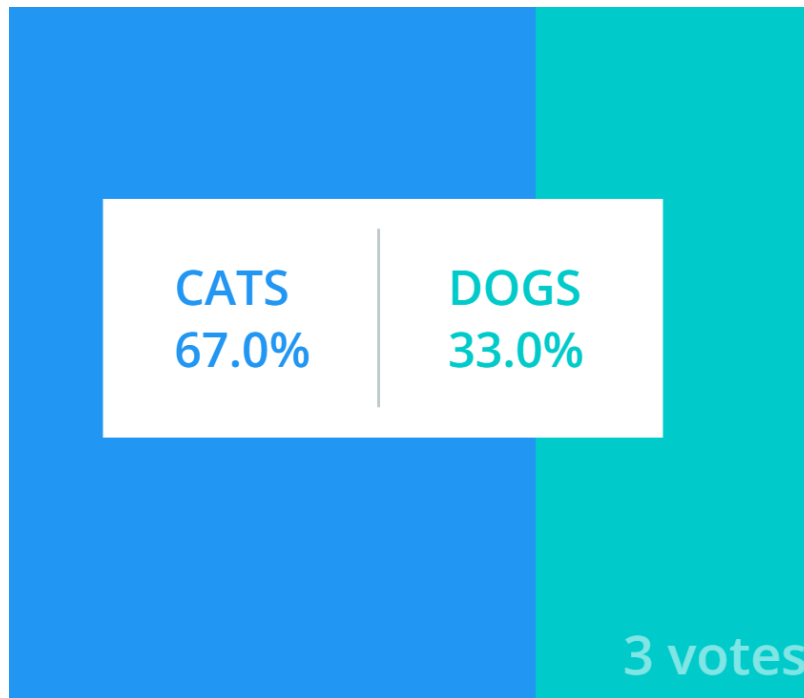


Restarted both both pods

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Clo
stributed Computing/project/New folder/example-voting-app (main)
$ kubectl rollout restart deployment vote -n default
deployment.apps/vote restarted

Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Clo
stributed Computing/project/New folder/example-voting-app (main)
$ kubectl rollout restart deployment result -n default
deployment.apps/result restarted
```

Data was same as before:



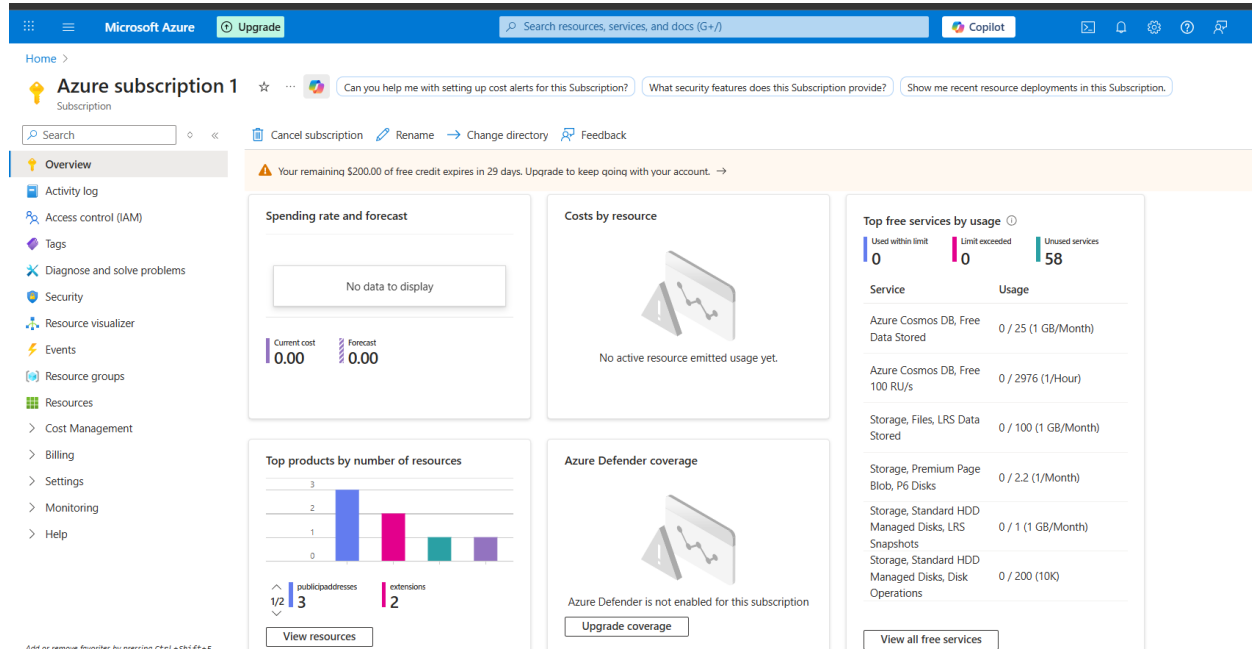
Database load efficiency

Despite heavy application load, database CPU utilization remained low due to effective caching and decoupled service design, indicating efficient data handling.

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.s
tributed Computing/project/New folder/example-voting-app (main)
$ kubectl top pods -n default
NAME                                CPU(cores)   MEMORY(bytes)
db-74574d66dd-hqk6m                 3m           31Mi
redis-6c5fb9c4b7-q7nd1              3m           5Mi
result-7df6cf5f8-28xds              1m           19Mi
result-7df6cf5f8-w4wqm              2m           20Mi
vote-77cc578dd-xmww8                1m           91Mi
vote-77cc578dd-z4nnb                1m           91Mi
worker-6f5f6cdd56-f88ww             14m          46Mi
```

Cost Efficiency Analysis

All of this done under my free 200\$ credits by azure bcz of the cost-effective methodologies that helped in cost optimization



My resources:

The screenshot shows the Microsoft Azure portal interface for 'Resource Manager | All resources'. The top navigation bar includes the Microsoft Azure logo, a search bar, and a Copilot icon. The left sidebar lists navigation options: Home, Resource Manager, and Default Directory. The main content area features a table of resources with columns for Name, Type, Resource Group, Location, and Subscription. The table is filtered by 'Subscription equals all', 'Resource Group equals all', 'Type equals all', and 'Location equals all'. The table shows a list of resources, including Public IP addresses, Network security groups, Virtual machines, Disks, Network interfaces, Virtual networks, Load balancers, and Network Watchers.

Name	Type	Resource Group	Location	Subscription
2295ef25-d62a-4c7c-a14e-1a41	Public IP address	MC_voting-app...	Southeast Asia	Azure subscript...
aks-agentpool-30049967-nsg	Network securit...	mc_voting-app...	Southeast Asia	Azure subscript...
aks-testnode-30486093-vms1	Virtual machine	MC_voting-app...	Southeast Asia	Azure subscript...
aks-testnode-30486093-vms1_C	Disk	MC_VOTING-A...	Southeast Asia	Azure subscript...
aks-testnode-30486093-vmsnic	Network Interfa...	MC_voting-app...	Southeast Asia	Azure subscript...
aks-vnet-30049967	Virtual network	MC_voting-app...	Southeast Asia	Azure subscript...
kubernetes	Load balancer	mc_voting-app...	Southeast Asia	Azure subscript...
kubernetes-a06387ca4d76849d1	Public IP address	mc_voting-app...	Southeast Asia	Azure subscript...
kubernetes-ab64cc9ccbc8e4093	Public IP address	mc_voting-app...	Southeast Asia	Azure subscript...
NetworkWatcher_southeastasia	Network Watcher	NetworkWatch...	Southeast Asia	Azure subscript...

Cost Optimization Strategies used

1) Autoscaling

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get hpa
NAME      REFERENCE                TARGETS      MINPODS  MAXPODS  REPLICAS  AGE
result    Deployment/result        cpu: <unknown>/50%    2        5        2         58m
vote      Deployment/vote          cpu: <unknown>/50%    2        5        2         62m
```

2) Stateless services

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get pvc
No resources found in default namespace.
```

Vote Service (Frontend - Python)

- No persistent storage in the container

Result Service (Frontend - Node.js)

- Reads data from PostgreSQL (external database)

Worker Service (Java)

- Processes votes from Redis queue
- Writes to PostgreSQL (external)

3) Smaller VM sizes

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Distributed Computing/project/New folder/example-voting-app (main)
$ kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
aks-testnode-30486093-vms1         Ready     <none>    3h6m    v1.33.5
```

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Compute infrastructure | Virtual machines >

aks-testnode-30486093-vms1 Virtual machine

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Connect
- Networking
- Settings
- Availability + scale
- Security
- Backup + disaster recovery
- Operations
- Monitoring
- Automation
- Help

Advisor (1 of 1): Enable Trusted Launch foundational excellence, and modern security for Existing Generation 2 VM(s)

Help me copy this VM in any region

Connect Start Restart Stop Hibernate

network/subnet

DNS name -

Size

Size Standard D2plds v5

vCPUs 2

RAM 4 GiB

Source image details

Source image -

publisher -

Source image offer -

Source image plan -

Disk

OS disk

aks-testnode-30486093-vms1_OsDisk_1_042d07dc285b40eba39f2443a6a8f657

4) Efficient Data Handling

Caching reduces database workloads

```
Personal@DESKTOP-2CLOJES MINGW64 /d/Arsalan khan/ARSALAN KHAN (UNI)/a.semester 7 fall 2025/Cloud and Di
stributed Computing/project/New folder/example-voting-app (main)
$ kubectl top pods -n default
NAME                                CPU(cores)   MEMORY(bytes)
db-74574d66dd-hqk6m                 3m           31Mi
redis-6c5fb9c4b7-q7nd1              3m           5Mi
result-7df6cfd5f8-28xds             1m           22Mi
result-7df6cfd5f8-w4wqm             1m           21Mi
vote-77cc578dd-xmww8               1m           91Mi
vote-77cc578dd-z4nnb               1m           91Mi
worker-6f5f6cdd56-f88ww             13m          46Mi
```

End..